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[5352]-168

S.E. (Computer Engg.) (II Sem.) EXAMINATION, 2018

MICROPROCESSOR & INTERFACING TECHNIQUES

(2012 PATTERN)

Time : Two Hours

Maximum Marks : 50

N.B. :— (i) Attempt total 4 questions : Q. No. 1 or Q. No. 2,
Q. No. 3 or Q. No. 4, Q. No. 5 or Q. No. 6,
Q. No. 7 or Q. No. 8.

(ii) Neat diagrams must be drawn wherever necessary.

(iii) Figures to the right indicate full marks.

1. (a) Compare 8086, 80386 processors on the basis of segmentation. [4]

(b) Explain the following addressing modes with suitable example
in 80386 processor : [4]

(i) Based mode

(ii) Indexed mode

(iii) Scaled indexed mode

(iv) Based scaled indexed mode.

(c) Explain the difference between DOS calls and BIOS calls. [4]

P.T.O.

Or

2. (a) Differentiate between Procedure and Macro. [4]
(b) Draw interfacing diagram between 8086 and 8259A. [4]
(c) Draw and discuss initialization sequence of 8259A. [4]
3. (a) Draw and explain mode set control word for 8255. What will be its value to initialize 8255 for the following configuration ? [6]
(i) Port B as mode 1 input
(ii) Port A as mode 0 output
(iii) Port C upper as input
(iv) Port C bit 2 as output.
- (b) Give salient features of 8237 DMA controller. [4]
(c) Define the following pins of ADC 0808 : [2]
(i) SOC
(ii) EOC.

Or

4. (a) Draw and explain internal block diagram of 8253 and prepare the control word for the following specifications : [6]
(i) BCD count
(ii) Counter 0
(iii) Mode 3
(iv) 16-bit count.

- (b) Draw and explain format of command word in 8279 for Keyboard/ Display Mode Set. [4]
- (c) What do you mean by NULL modem ? Explain it with neat diagram. [2]
5. (a) What is the difference between minimum and maximum mode of 8086 ? [3]
- (b) Draw 8086 based minimum mode system showing 4×4 matrix keyboard using 8255. [6]
- (c) Explain the role of the following chips in maximum mode configuration : [4]
- (i) 8288
- (ii) 8286.
- Or*
6. (a) What is difference between memory mapped I/O and I/O mapped I/O ? [4]
- (b) Explain the following 8087 instructions with *one* example each : [6]
- (i) FMUL
- (ii) FDIV
- (iii) FBST.
- (c) Explain 32-bit single precision format for floating point numbers. [3]

7. (a) Explain the following Intel 82801 1JR I/O controller Hub capabilities : [6]
- (i) Direct Media Interface
 - (ii) PCI Express Interface
 - (iii) Serial ATA (SATA) controller.
- (b) Draw and explain block diagram of Intel core i5 processor. [7]

Or

8. (a) Explain features of Intel X58 Chipset. [5]
- (b) Explain blocks of Intel i5 motherboard : [8]
- (i) System memory
 - (ii) Graphics system
 - (iii) USB
 - (iv) LAN subsystem.